

Databricks Lab : Spark Broadcast Join Optimization

Duration : 60 Min.

Scenario

Optimize big data joins by broadcasting a small metadata table to worker nodes, preventing shuffle overhead.

Input Details

The environment has been pre-configured with the following resources: - **Workspace:** Pre-configured through KodeBuck Service credentials (DATABRICKS_HOST, DATABRICKS_TOKEN)

Username & Naming Conventions

Your candidate prefix is a combination of your username (the part of your email/login before the @ or _ symbol) and exam code. For example, if your username is student (or student_labskraft.com) and your exam code is exam123, your prefix is **student-exam123**.

You must name your resources accordingly:

- Optimized Table:** <prefix>_broadcast_join (e.g. student_exam123_broadcast_join)

Task Objectives

1. Implement Broadcast Join

- Write a query broadcasting the smaller table.
- Confirm the broadcast hint in the query physical execution plan.

Verification

Once you have performed the tasks, you can run the verification script to check your progress and receive your score. The verification system will check the actual resources in the environment.

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Join outcome correctness	4 Marks
TC2	Broadcast hint check in AST	4 Marks

TC3	Optimization execution check	4 Marks
TC4	Query plan metrics validation	4 Marks
TC5	Output schema structure check	4 Marks

Total Score: 20 Marks

Important Notes

- Ensure all resource names contain your candidate username prefix in lowercase.
- Check that your script compiles cleanly and contains no syntax errors.